

1.Exercises:

Requirement: Students can enter marks from 0 to 100.

Identify the equivalence partitions for the marks input field. Divide the possible inputs into valid and

Test Case	Equivalence Class	Type	Test Value	Expected Result
TC01	Less than 0	Invalid	-1	Reject
TC02	0 to 100	Valid	0	Accept
TC03	0 to 100	Valid	50	Accept
TC04	0 to 100	Valid	100	Accept
TC05	Greater than 100	Invalid	101	Reject
TC06	Negative decimal	Invalid	-10.5	Reject
TC07	Greater than 100 decimal	Invalid	100.5	Reject

2. Exercise – ATM Withdrawal Validation

Identify all the equivalence partitions for the withdrawal amount and provide one suitable test value

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Amount < 0	Invalid	-500	Reject
TC02	Amount = 0	Invalid	0	Reject
TC03	0 < Amount ≤ Balance	Valid	500	Accept
TC04	Amount = Available Balance	Valid	10,000	Accept
TC05	Amount > Available Balance	Invalid	10,001	Reject
TC06	Decimal amount	Valid/Invalid*	500.5	Based on requirement
TC07	Non-numeric amount	Invalid	ABC	Reject

Exercise 3 – Username Validation

Requirement: Username must contain **5 to 15 characters**.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than 5 characters	Invalid	abc (3 chars)	Reject
TC02	Exactly 5 characters	Valid	abcde	Accept
TC03	Between 5 and 15 characters	Valid	username1 (9 chars)	Accept

TC04	Exactly 15 characters	Valid	abcdefghijklmno	Accept
TC05	More than 15 characters	Invalid	abcdefghijklmno p (16 chars)	Reject

Exercise 4 – Mobile Number Validation

Requirement: Mobile number must contain **exactly 10 digits**.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than 10 digits	Invalid	987654321	Reject
TC02	Exactly 10 digits	Valid	9876543210	Accept
TC03	More than 10 digits	Invalid	98765432101	Reject
TC04	No digits / Empty	Invalid	" Empty"	Reject
TC05	Contains letters	Invalid	98765ABCDE	Reject
TC06	Contains special characters	Invalid	98765@3210	Reject
TC07	Contains spaces	Invalid	98765 43210	Reject

Exercise 5 – Shopping Cart Quantity

Requirement: A customer can purchase **1 to 10 items**.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than 1	Invalid	0	Reject
TC02	1 to 10 items	Valid	5	Accept
TC03	Exactly 1 item	Valid	1	Accept
TC04	Exactly 10 items	Valid	10	Accept
TC05	Greater than 10	Invalid	11	Reject
TC06	Negative quantity	Invalid	-1	Reject
TC07	Non-numeric input	Invalid	ABC	Reject

Exercise 6 – Bank Account Balance

Requirement: Minimum balance = **₹1,000** and maximum balance = **₹10,00,000**.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than ₹1,000	Invalid	₹ 500	Reject
TC02	Exactly ₹1,000	Valid	₹ 1,000	Accept
TC03	Between ₹1,000 and ₹10,00,000	Valid	₹ 50,000	Accept

TC04	Exactly ₹10,00,000	Valid	₹ 10,00,000	Accept
TC05	Greater than ₹10,00,000	Invalid	₹ 10,00,001	Reject
TC06	Negative balance	Invalid	₹ -500	Reject
TC07	Non-numeric input	Invalid	ABC	Reject

Exercise 7 – Employee Salary

Requirement: Salary must be between ₹15,000 and ₹2,00,000.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than ₹15,000	Invalid	₹ 10,000	Reject
TC02	Exactly ₹15,000	Valid	₹ 15,000	Accept
TC03	Between ₹15,000 and ₹2,00,000	Valid	₹ 50,000	Accept
TC04	Exactly ₹2,00,000	Valid	₹ 2,00,000	Accept
TC05	Greater than ₹2,00,000	Invalid	₹ 2,50,000	Reject
TC06	Negative salary	Invalid	₹ -5,000	Reject
TC07	Non-numeric input	Invalid	ABC	Reject

Exercise 8 – Movie Ticket Booking

Requirement: A customer can book 1–6 tickets per transaction.

Test Case	Equivalence Partition	Type	Test Value	Expected Result
TC01	Less than 1 ticket	Invalid	0	Reject
TC02	Exactly 1 ticket	Valid	1	Accept
TC03	Between 1–6 tickets	Valid	3	Accept
TC04	Exactly 6 tickets	Valid	6	Accept
TC05	More than 6 tickets	Invalid	7	Reject
TC06	Negative quantity	Invalid	-1	Reject
TC07	Non-numeric input	Invalid	ABC	Reject

1. Requirement for the application

Suppose we have a system:

Input: A square matrix

2	1	0	0
0	5	2	0
0	0	7	3

Process: Calculate eigenvalues and eigenvectors and determine whether the matrix can be diagonalized.
Output: If diagonalizable, display matrices P and D such that

TEST CASE ID	INPUT	TEST CASE	TEST VALUE
TC-01	Square Matrix	Valid	[[2,1,0,0],[0,5,2,0],[0,0,7,3],[0,0,0,9]]
TC-02	Square Matrix (Diagonalizable)	Valid	[[2,1],[0,3]]
TC-03	Non-Square Matrix	Invalid	[[1,2,3],[4,5,6]]
TC-04	Empty Matrix	Invalid	[]
TC-05	Non-numeric Input	Invalid	[["a","b"],["c","d"]]

and invalid equivalence classes. Provide one suitable test value for each partition.

Partition	Range	Valid/Invalid
P1	Marks < 0	Invalid
P2	Marks 0–100	Valid
P3	Marks > 100	Invalid

Partition	Range	Valid/Invalid
P1	Amount < ₹500	Invalid
P2	₹500–₹20,000	Valid
P3	Amount > ₹20,000	Invalid

Partition	Range	Valid/Invalid
P1	Less than 5 characters	Invalid
P2	5–15 characters	Valid

P3	More than 15 characters	Invalid
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Partition	Range	Valid/Invalid
P1	Less than 10 digits	Invalid
P2	Exactly 10 digits	Valid
P3	More than 10 digits	Invalid

Partition	Range	Valid/Invalid
P1	Quantity < 1	Invalid
P2	Quantity 1–10	Valid
P3	Quantity > 10	Invalid

Partition	Range	Valid/Invalid
P1	Balance < ₹1,000	Invalid
P2	₹1,000–₹10,00,000	Valid

P3	Balance > ₹10,00,000	Invalid
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Partition	Range	Valid/Invalid
P1	Salary < ₹15,000	Invalid
P2	₹15,000–₹2,00,000	Valid
P3	Salary > ₹2,00,000	Invalid

Partition	Range	Type
P1	Less than 1 ticket	Invalid
P2	1–6 tickets	Valid
P3	More than 6 tickets	Invalid

alized.

PARTITION	RANGE / CONDITION	VALID/INVALID
P1	Square matrix and diagonalizable	Valid
P2	Square matrix but not diagonalizable	Valid
P3	Non-square matrix	Invalid
P4	Empty matrix	Invalid

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